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1. (Special note: Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file [here](#) . If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)
- 1 / 1 point

For each of the following sets, select all that are convex:

☒ $S = \{(x_1, x_2) \in \mathbb{R}^2 | x_1^2 + x_2^2 \leq 4, x_1 \geq 0, x_2 \leq 0\}.$

Correct

☐ $S = \{(x_1, x_2) \in \mathbb{Z}^2 | x_1^2 + x_2^2 \leq 4, x_1 \geq 0, x_2 \leq 0\}.$

☒ $S = \{x \in \mathbb{R}^n | \sum_{i=1}^n x_i^2 \leq 4, x_1 \geq 0, x_2 \leq 0\}.$

Correct

☒ $S = \{x \in \mathbb{R}^n | x_1 + x_2 = 10(x_3 + x_4)\}, where n \geq 4.$

Correct

☐ $S = \{x \in \mathbb{R}^n | \max_{i=1,...,n} \{x_i\} = 10 \min_{i=1,...,n} \{x_i\}\}.$

2. For each of the following functions, select all that are convex over the given region:
- 1 / 1 point

☐ $f(x) = 2x^3 - x^2 - 2x + 1$ for $x \in \mathbb{R}.$

☒ $f(x) = x^2 - 2x + 1$ for $x \in \mathbb{R}.$

Correct

☐ $f(x) = x_1^2 - 4x_1x_2 + 3x_2^2 + 3x_1 + 4x_2$ for $x \in \mathbb{R}^2.$

☒ $f(x) = \sum_{i=1}^n (x_i - a)^2$, where $a > 0$ is a given parameter, for $x \in \mathbb{R}^n.$

Correct

☒ $f(\alpha, \beta) = \sum_{i=1}^n (\alpha + \beta x_i - y_i)^2$, where $x_i \in \mathbb{R}$ and $y_i \in \mathbb{R}$ are given parameters, $i = 1, \dots, n$, for $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{R}.$

Correct

3. For the nonlinear program
- 1 / 1 point

$$\begin{array}{ll} \min & 2x^3 - x^2 - 2x + 1 \\ \text{s.t.} & x \geq -1, \end{array}$$

which of the following statements is correct?

- ☐ This is a convex program.
- ☒ There are two local minimizers.
- ☐ The unique global minimizer is a boundary point.
- ☐ This program is unbounded.
- ☐ None of the above.

Correct

4. For each of the following matrices, select all that are positive semi-definite:
- 1 / 1 point

☒ $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$

Correct

☐ $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}.$

☒ $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}.$

Correct

☒ $A = \begin{bmatrix} 1 & 2 & 3 & \cdots & n \\ 0 & 2 & 3 & \cdots & n \\ 0 & 0 & 3 & \cdots & n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & n \end{bmatrix}$ for some $n \geq 3.$

Correct

☐ $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$

5. For the following statements, select all that are incorrect:
- 1 / 1 point

- ☐ An optimal solution to a linear program must be a boundary point.
- ☒ An optimal solution to a linear program must be an extreme point.

Correct

- ☐ An optimal solution to a nonlinear program can be an interior point.
- ☒ An optimal solution to a nonlinear program must be an interior point.

Correct

- ☐ A global optimal solution to a nonlinear program must be a local optimal solution.